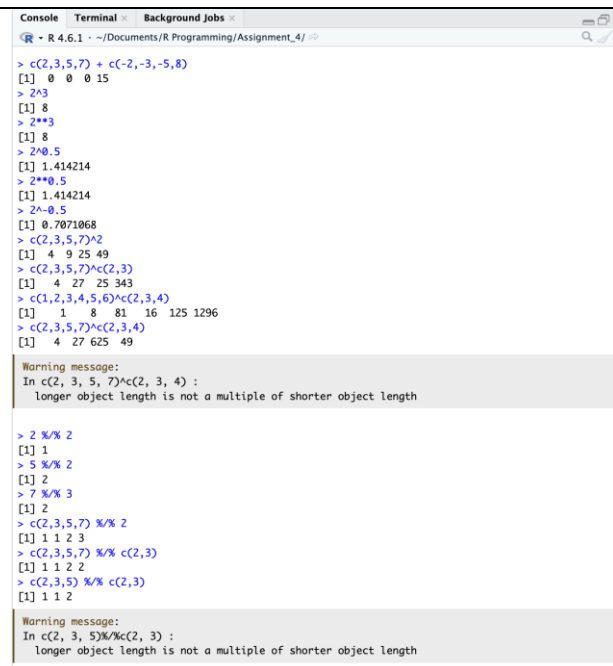
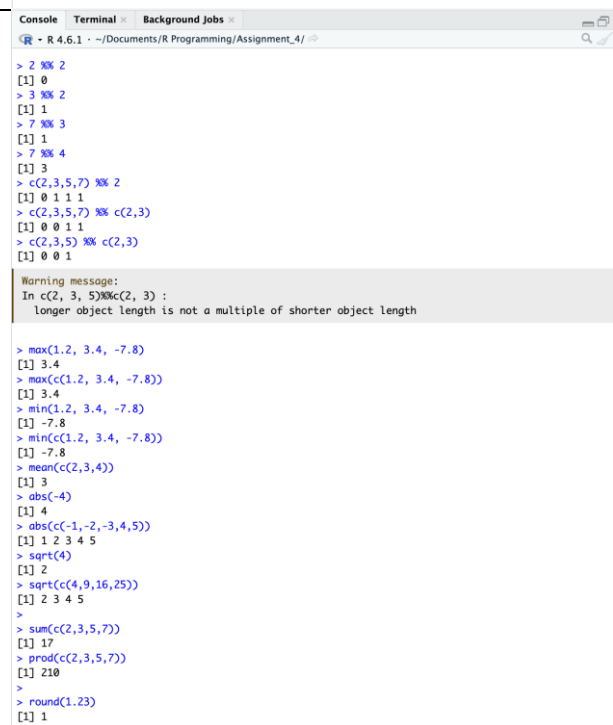


|            |                    |
|------------|--------------------|
| Name :     | Siddhanth Mungekar |
| Roll No    | 23102B0030         |
| Branch     | CM PN/B            |
| NPTEL Week | 3                  |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Screenshots |  <pre> R 4.6.1 - ~/Documents/R Programming/Assignment_4/ &gt; c(2,3,5,7) + c(-2,-3,-5,8) [1] 0 0 0 15 &gt; 2^3 [1] 8 &gt; 2**3 [1] 8 &gt; 2^0.5 [1] 1.414214 &gt; 2**0.5 [1] 1.414214 &gt; 2^-0.5 [1] 0.7071068 &gt; c(2,3,5,7)^2 [1] 4 9 25 49 &gt; c(2,3,5,7)^c(2,3) [1] 4 27 25 343 &gt; c(1,2,3,4,5,6)^c(2,3,4) [1] 1 8 81 16 125 1296 &gt; c(2,3,5,7)^c(2,3,4) [1] 4 27 625 49  Warning message: In c(2, 3, 5, 7)^c(2, 3, 4) : longer object length is not a multiple of shorter object length  &gt; 2 %/% 2 [1] 1 &gt; 5 %/% 2 [1] 2 &gt; 7 %/% 3 [1] 2 &gt; c(2,3,5,7) %/% 2 [1] 1 1 2 3 &gt; c(2,3,5,7) %/% c(2,3) [1] 1 1 2 2 &gt; c(2,3,5) %/% c(2,3) [1] 1 1 2  Warning message: In c(2, 3, 5)%/%c(2, 3) : longer object length is not a multiple of shorter object length </pre> |
|             |  <pre> R 4.6.1 - ~/Documents/R Programming/Assignment_4/ &gt; 2 %% 2 [1] 0 &gt; 3 %% 2 [1] 1 &gt; 7 %% 3 [1] 1 &gt; 7 %% 4 [1] 3 &gt; c(2,3,5,7) %% 2 [1] 0 1 1 1 &gt; c(2,3,5,7) %% c(2,3) [1] 0 0 1 1 &gt; c(2,3,5) %% c(2,3) [1] 0 0 1  Warning message: In c(2, 3, 5)%%c(2, 3) : longer object length is not a multiple of shorter object length  &gt; max(1.2, 3.4, -7.8) [1] 3.4 &gt; max(c(1.2, 3.4, -7.8)) [1] 3.4 &gt; min(1.2, 3.4, -7.8) [1] -7.8 &gt; min(c(1.2, 3.4, -7.8)) [1] -7.8 &gt; mean(c(2,3,4)) [1] 3 &gt; abs(-4) [1] 4 &gt; abs(c(-1,-2,-3,4,5)) [1] 1 2 3 4 5 &gt; sqrt(4) [1] 2 &gt; sqrt(c(4,9,16,25)) [1] 2 3 4 5 &gt; &gt; sum(c(2,3,5,7)) [1] 17 &gt; prod(c(2,3,5,7)) [1] 210 &gt; &gt; round(1.23) [1] 1 </pre>                                             |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <div> <div>ConsoleTerminalBackground Jobs</div> <div> R • R 4.6.1 • ~/Documents/R Programming/Assignment_4/ </div> </div> <pre> &gt; round(1.83) [1] 2 &gt; &gt; log(10) [1] 2.302585 &gt; log(exp(1)) [1] 1 &gt; log(c(10,100,1000)) [1] 2.302585 4.605170 6.907755 &gt; &gt; log10(10) [1] 1 &gt; log10(100) [1] 2 &gt; log10(c(10,100,1000)) [1] 1 2 3 &gt; x1 = c(1,2,3,4) &gt; x1 [1] 1 2 3 4 &gt; &gt; x2 = x1^2 &gt; x2 [1] 1 4 9 16 &gt; c(1,2,3,4) + sum(c(1,2,3,4)) * prod(c(1,2)) [1] 21 22 23 24 &gt; &gt; abs(c(1,2,3,4) - sum(c(1,2,3,4)) * prod(c(1,2))) [1] 19 18 17 16 &gt; x = matrix( +   nrow = 4, +   ncol = 2, +   data = c(1,2,3,4,5,6,7,8) + ) &gt; x       [,1] [,2] [1,]    1    5 [2,]    2    6 [3,]    3    7 [4,]    4    8 &gt; x[3,2] [1] 7 </pre> |
|  | <div> <div>ConsoleTerminalBackground Jobs</div> <div> R • R 4.6.1 • ~/Documents/R Programming/Assignment_4/ </div> </div> <pre> &gt; x = matrix( +   nrow = 4, +   ncol = 2, +   data = c(1,2,3,4,5,6,7,8), +   byrow = FALSE + ) &gt; x       [,1] [,2] [1,]    1    5 [2,]    2    6 [3,]    3    7 [4,]    4    8 &gt; y = matrix( +   nrow = 4, +   ncol = 2, +   data = c(1,2,3,4,5,6,7,8), +   byrow = TRUE + ) &gt; y       [,1] [,2] [1,]    1    2 [2,]    3    4 [3,]    5    6 [4,]    7    8 &gt; dim(x) [1] 4 2 &gt; nrow(x) [1] 4 &gt; ncol(x) [1] 2 &gt; mode(x) [1] "numeric" &gt; attributes(x) \$dim [1] 4 2 </pre>                                                                                                                                              |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |                        |  |   |                                |  |   |                                |  |        |  |  |    |                   |  |    |                    |  |  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------------|--|---|--------------------------------|--|---|--------------------------------|--|--------|--|--|----|-------------------|--|----|--------------------|--|--|
|              | <pre>&gt; help("matrix") &gt; is.matrix(as.matrix(1:10)) [1] TRUE &gt; data(warpbreaks) &gt; !is.matrix(warpbreaks) [1] TRUE &gt; warpbreaks[1:10,]   breaks wool tension 1     26    A      L 2     30    A      L 3     54    A      L 4     25    A      L 5     70    A      L 6     52    A      L 7     51    A      L 8     26    A      L 9     67    A      L 10    18    A      M &gt; &gt; as.matrix(warpbreaks[1:10,])   breaks wool tension 1  "26"  "A"  "L" 2  "30"  "A"  "L" 3  "54"  "A"  "L" 4  "25"  "A"  "L" 5  "70"  "A"  "L" 6  "52"  "A"  "L" 7  "51"  "A"  "L" 8  "26"  "A"  "L" 9  "67"  "A"  "L" 10 "18"  "A"  "M" &gt;  </pre>                                                                                                                                                                              |              |                        |  |   |                                |  |   |                                |  |        |  |  |    |                   |  |    |                    |  |  |
|              | <div><div>FilesPlotsPackagesHelpViewerPresentation</div><div><div>R: Matrices - Find in Topic</div><div><div>matrix (base)R Documentation</div><div><h2>Matrices</h2><p><b>Description</b></p><p><code>matrix</code> creates a matrix from the given set of values.</p><p><code>as.matrix</code> attempts to turn its argument into a matrix.</p><p><code>is.matrix</code> tests if its argument is a (strict) matrix.</p><p><b>Usage</b></p><pre>matrix(data = NA, nrow = 1, ncol = 1, byrow = FALSE,         dimnames = NULL)  as.matrix(x, ...) ## S3 method for class 'data.frame' as.matrix(x, rownames.force = NA, ...)  is.matrix(x)</pre><p><b>Arguments</b></p><p><code>data</code> an optional data vector (including a list or <code>expression</code> vector). Non-atomic classes of R objects</p></div></div></div></div> |              |                        |  |   |                                |  |   |                                |  |        |  |  |    |                   |  |    |                    |  |  |
|              | <div><div>Data</div><table><tr><td>ⓘ warpbreaks</td><td>54 obs. of 3 variables</td><td></td></tr><tr><td>x</td><td>num [1:4, 1:2] 1 2 3 4 5 6 7 8</td><td></td></tr><tr><td>y</td><td>num [1:4, 1:2] 1 3 5 7 2 4 6 8</td><td></td></tr><tr><td colspan="3">Values</td></tr><tr><td>x1</td><td>num [1:4] 1 2 3 4</td><td></td></tr><tr><td>x2</td><td>num [1:4] 1 4 9 16</td><td></td></tr></table></div>                                                                                                                                                                                                                                                                                                                                                                                                                               | ⓘ warpbreaks | 54 obs. of 3 variables |  | x | num [1:4, 1:2] 1 2 3 4 5 6 7 8 |  | y | num [1:4, 1:2] 1 3 5 7 2 4 6 8 |  | Values |  |  | x1 | num [1:4] 1 2 3 4 |  | x2 | num [1:4] 1 4 9 16 |  |  |
| ⓘ warpbreaks | 54 obs. of 3 variables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |                        |  |   |                                |  |   |                                |  |        |  |  |    |                   |  |    |                    |  |  |
| x            | num [1:4, 1:2] 1 2 3 4 5 6 7 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |                        |  |   |                                |  |   |                                |  |        |  |  |    |                   |  |    |                    |  |  |
| y            | num [1:4, 1:2] 1 3 5 7 2 4 6 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |                        |  |   |                                |  |   |                                |  |        |  |  |    |                   |  |    |                    |  |  |
| Values       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |              |                        |  |   |                                |  |   |                                |  |        |  |  |    |                   |  |    |                    |  |  |
| x1           | num [1:4] 1 2 3 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |              |                        |  |   |                                |  |   |                                |  |        |  |  |    |                   |  |    |                    |  |  |
| x2           | num [1:4] 1 4 9 16                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |                        |  |   |                                |  |   |                                |  |        |  |  |    |                   |  |    |                    |  |  |